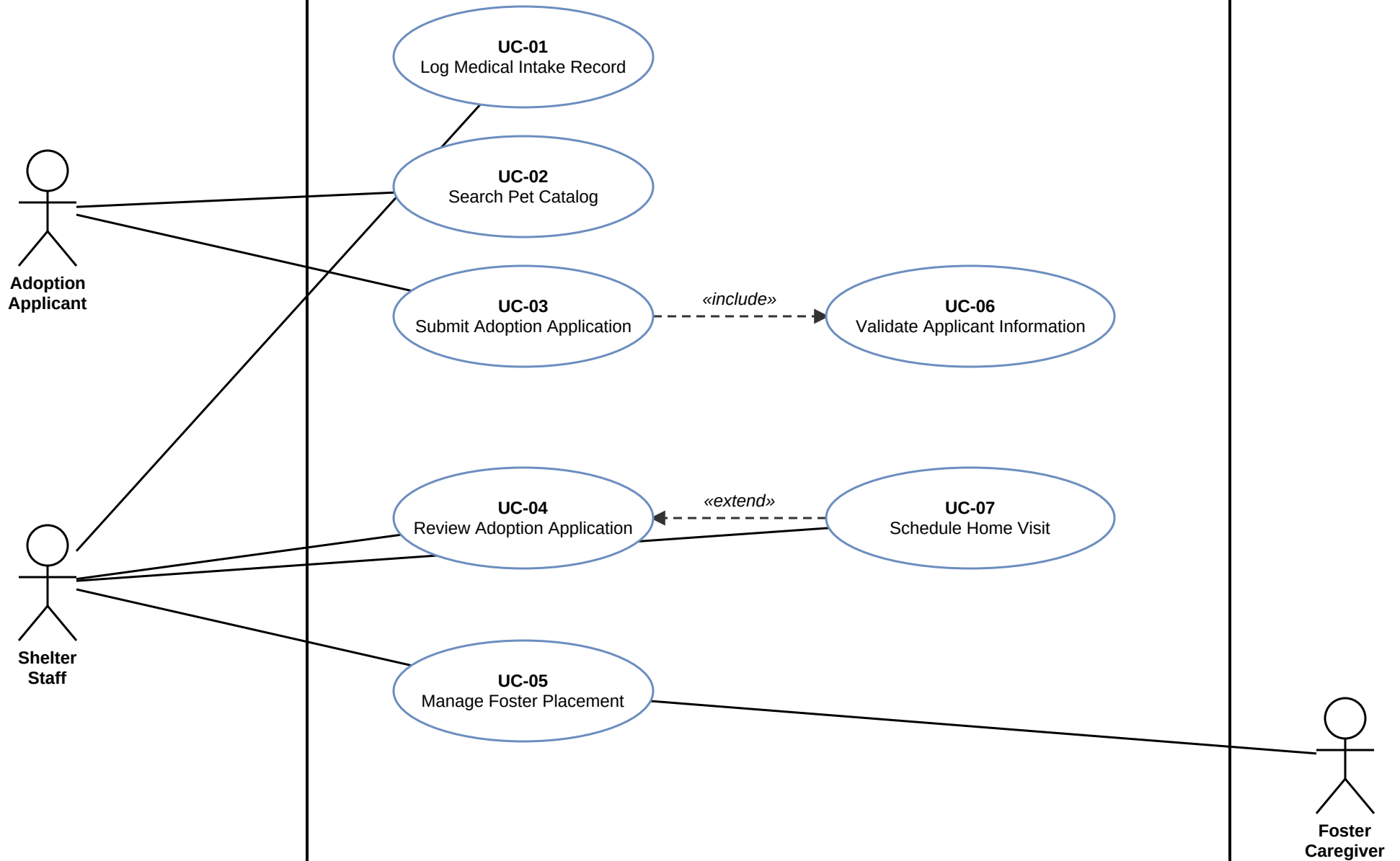


Pet Adoption & Shelter Management System



Problem #57 — Pet Adoption & Shelter Management System — Requirements Table

Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow staff to log animal vaccination and medical intake records, updating animal availability status to 'Ready for Adoption'.	High	Pass: Adoption application linked to verified animal record; Fail: Animal under medical quarantine marked adopted.	Given by problem statement. Core safeguard ensuring only medically cleared animals can be adopted.
FR-002	Functional	The system shall allow an adoption applicant to search and filter the pet catalog by Species, Age, Breed, and Good-with-Kids.	High	Pass: Applying any single filter or combination returns only animals matching all selected criteria, with zero mismatched results in 10 test queries; Fail: Any returned animal does not match the applied filter(s).	Directly supports NFR-001 (catalog performance) and lets applicants find suitable pets quickly, increasing adoption rate.
FR-003	Functional	The system shall allow an adoption applicant to submit an adoption questionnaire/application for a specific animal listed as 'Ready for Adoption'.	High	Pass: Submitted application is stored with applicant details, selected animal ID, and a 'Submitted' timestamp visible to staff; Fail: Application can be submitted for an animal not marked 'Ready for Adoption', or is not saved.	Captures the adopter-screening data (household, experience, home environment) shelter staff need to evaluate suitability.
FR-004	Functional	The system shall allow shelter staff to review a submitted adoption application and record a decision of Approved or Rejected with a reason.	Medium	Pass: Every application transitions to exactly one of 'Approved' or 'Rejected' with a non-empty reason field recorded; Fail: A decision is saved without a reason, or an application remains undecided after staff submits a decision.	Ensures screening decisions are documented and auditable, protecting animal welfare and applicant accountability.
FR-005	Functional	The system shall allow shelter staff to assign an approved animal to a foster home and record the foster placement start date and scheduled review/return date.	Medium	Pass: A foster placement record stores animal ID, foster home ID, start date, and review date, and the animal's status updates to 'In Foster Care'; Fail: A placement is created without a review date, or animal status does not update.	Supports management of foster home placement timelines, a core function named in the problem context.
NFR-001	Nonfunctional (Performance & Security)	The pet search catalog must support multi-attribute filtering (Species, Age, Breed, Good with Kids) with page response < 200 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given by problem statement. Slow or insecure search would frustrate applicants and risk exposing shelter data.
NFR-002	Nonfunctional (Security & Privacy)	The system shall encrypt adoption applicants' personal and contact information, both at rest and in transit, and restrict access to authenticated shelter staff only.	High	Pass: A security scan confirms applicant data fields are encrypted in the database and over HTTPS, and an unauthenticated or non-staff request to view applicant records is rejected in 100% of test attempts; Fail: Applicant data is readable in plaintext at rest, in	Adoption questionnaires collect sensitive personal data (household details, contact info); protecting it is required for applicant trust and data-protection compliance.

Use-Case Flow

Pet Adoption & Shelter Management System

Use Case ID: UC-03

Use Case Name: Submit Adoption Application

Primary Actor: Adoption Applicant

Preconditions

1. Applicant is logged into the system.
2. The animal status is "Ready for Adoption".
3. Applicant found the animal using catalog search.

Postconditions

1. Application is saved with status "Submitted".
2. Shelter staff can see the application for review.

Main Success Scenario

1. Applicant selects an animal and clicks "Apply to Adopt".
2. System shows the adoption form.
3. Applicant fills in the form with personal and household details.
4. Applicant submits the form.
5. System checks that all fields are filled correctly.
6. System saves the application and marks it as "Submitted".
7. System shows a confirmation message to the applicant.
8. Use case ends.

Alternate Flow: Invalid or Incomplete Information

- 5a1. System finds a missing or wrong field.
- 5a2. System shows an error message.
- 5a3. Applicant corrects the field and submits again.